Parametric models and logistic regression

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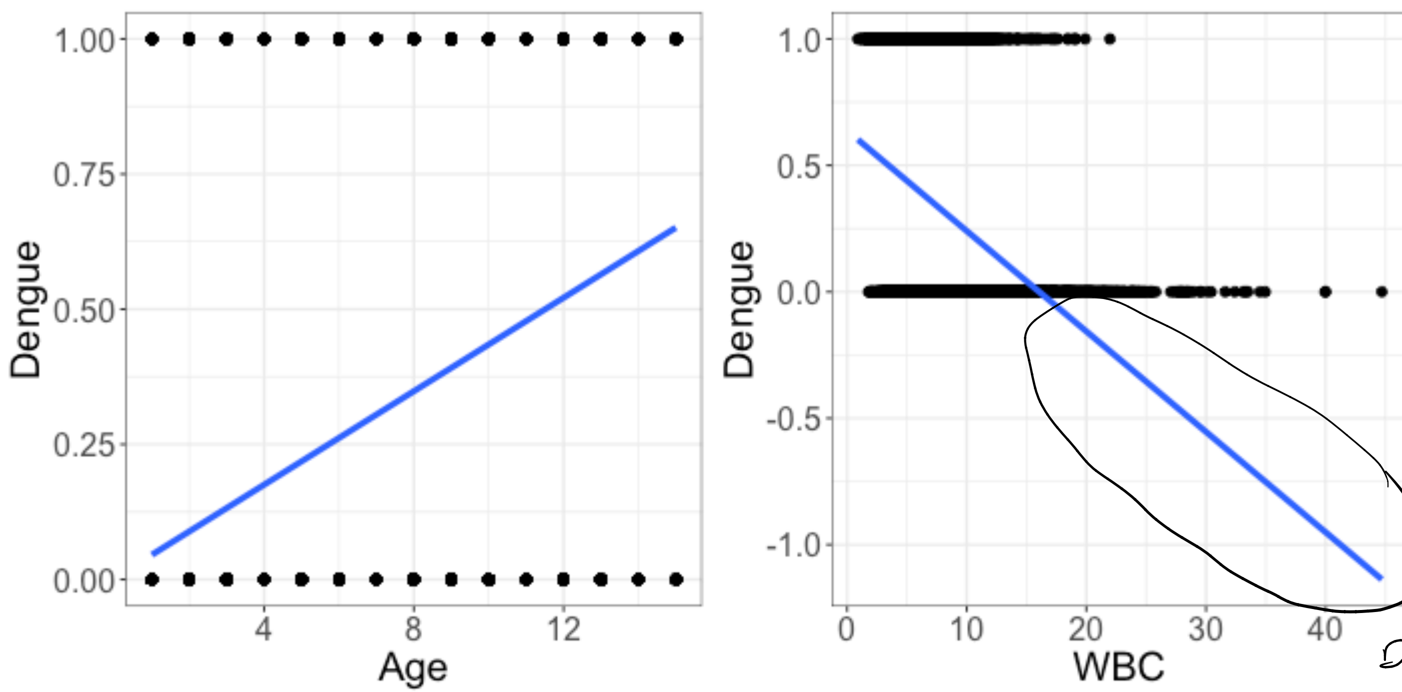
Last time

Dengue $\in \{0, 1\}$

Don't fit linear regression with a binary response



$\hat{\text{Dengue}} < 0$

??

$\hat{\text{Dengue}} = 0.6$?

Want: model probabilities

$\Rightarrow$ need predictions to stay between 0 & 1

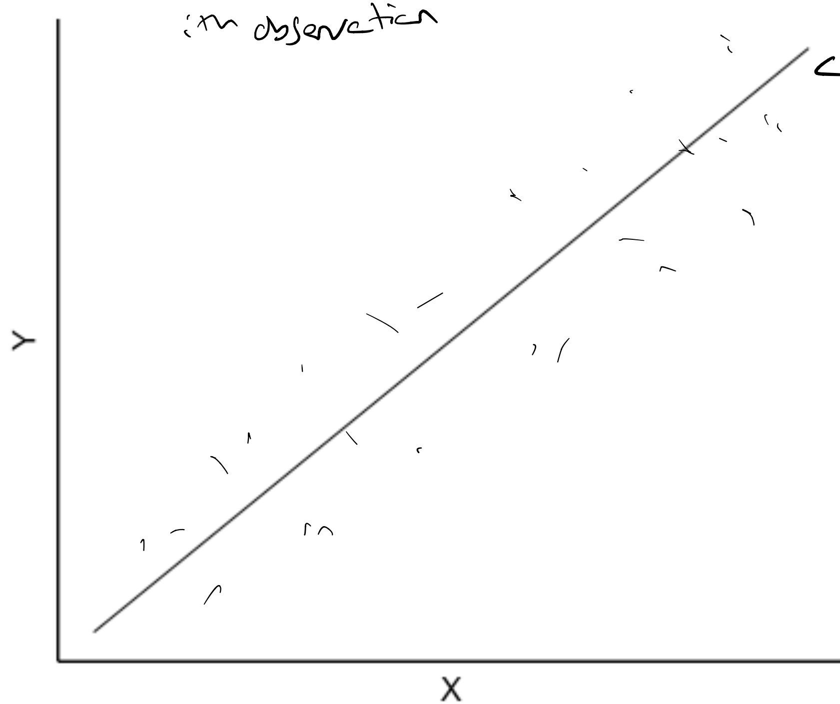
$i = 1, \ldots, n$ (n = sample size)

Revisiting the linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

variability around the line

ith observation



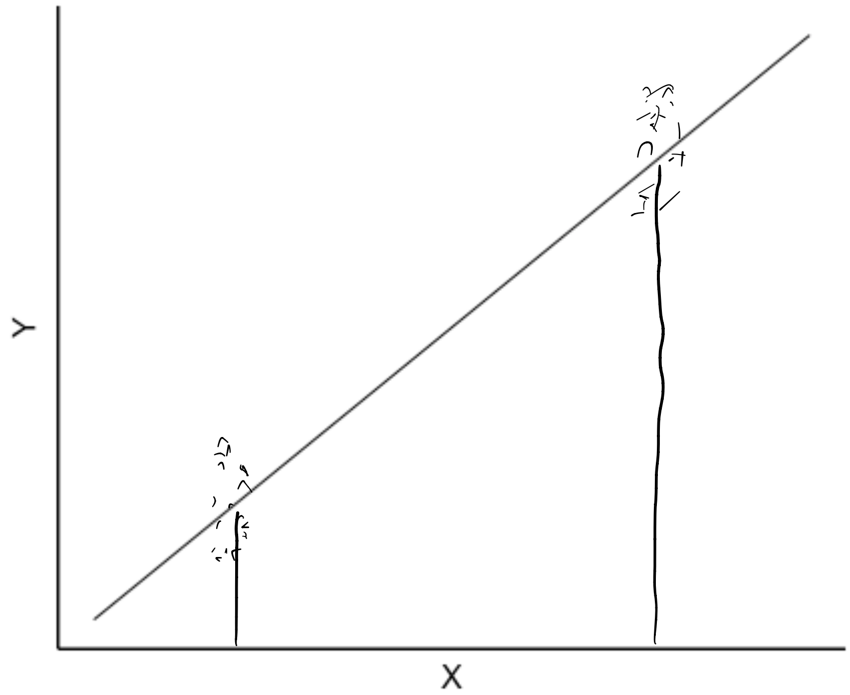
general (average) trend for Y vs X

$\beta_0 + \beta_1 X$

> Will all of the observations fall exactly on the line?

Revisiting the linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$



> Given a value of X, how do I know where the values of Y are likely to be?

iid = independent and identically distributed

Revisiting the linear regression model

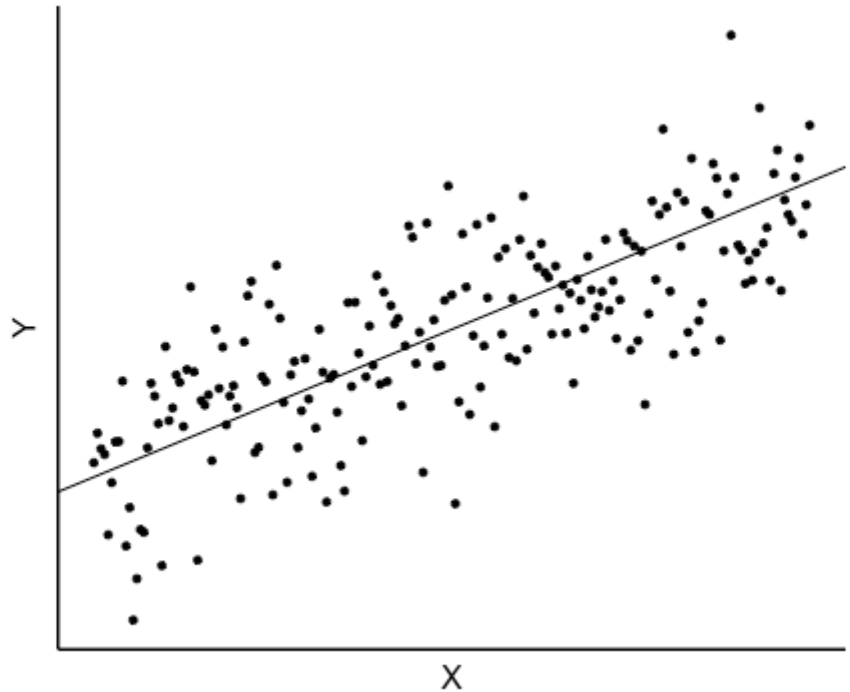
$\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

normal

mean 0

constant variance

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$



What do we often assume about the distribution of ε?

Revisiting the linear regression model

$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \qquad \varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

$\Rightarrow$ Y_i is <u>also</u> normal

$Y_i \sim N(\underbrace{\beta_0 + \beta_1 X_i}_{\text{line}}, \sigma^2)$ ← variability around line

(<u>average</u> value of Y, given X)

$$\begin{cases} Y_i \sim N(\mu_i, \sigma^2) \\ \mu_i = \beta_0 + \beta_1 X_i \end{cases}$$

mean → μ_i, variance → σ^2

← specifies distribution of Y

← relate distribution to X

Parametric modeling

A regression model is an example of a more general process called **parametric modeling**

- **Step 1:** Choose a reasonable distribution for Y_i e.g. $Y_i \sim N(\mu_i, \sigma^2)$
- **Step 2:** Build a model for the parameters of interest $\mu_i = \beta_0 + \beta_1 X_i$
- **Step 3:** Fit the model

e.g. lm(..) in R

Step 1: Choose a reasonable distribution for Y_i

> What do I mean by a *distribution*?

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- A **distribution** tells us what outcomes are possible for Y_i, and how often these outcomes occur.

Here the possible values of Y_i are 0 (no dengue) and 1 (dengue).

> How often do these values occur in the population?

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> How often do these values occur in the population?

- We don't know, so we will estimate from the sample
- We assume the probability $Y_i = 1$ depends on Age_i

Step 1: Choose a reasonable distribution for Y_i

How should I describe the distribution of Y_i?

$Y_i = 0$ or $Y_i = 1$

$\pi_i = P(Y_i = 1)$ (probability an individual has dengue)

$\Rightarrow P(Y_i = 0) = 1 - \pi_i$

That's it!

we have one parameter to estimate (π_i)

we want π_i to depend on variables like Age_i

Bernoulli distribution

Definition: Let Y_i be a binary random variable, and $\pi_i = P(Y_i = 1)$. Then $Y_i \sim Bernoulli(\pi_i)$.

> What do I mean by a *random variable*?

Bernoulli distribution

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> What do I mean by a *random variable*?

A **random variable** is an event that has a set of possible outcomes, but we don't know which one will occur

- Here $Y_i = 0$ or 1
- Our goal is to use the observed data to estimate $\pi_i = P(Y_i = 1)$

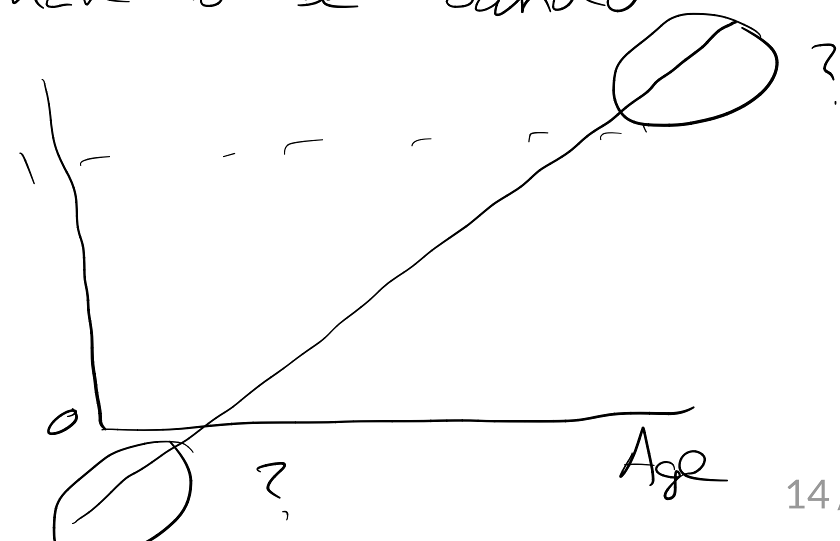
Second attempt at a model

$$Y_i \sim Bernoulli(\pi_i) \qquad \pi_i = P(Y_i = 1 | Age_i)$$

$$\pi_i = \beta_0 + \beta_1 Age_i$$

> Are there still any potential issues with this approach?

$\pi_i \in [0,1]$ but $\beta_0 + \beta_1 Age_i$ does not have to be banded



Fixing the issues

$\log(x) < 0 \quad x < 1$

$\log(1) = 0$

$\log(x) > 0 \quad \text{if } x > 1$

$Y_i \sim \text{Bernoulli}(\pi_i)$

$$g(\pi_i) = \underbrace{\beta_0 + \beta_1 Age_i}_{\in (-\infty, \infty)}$$

want $\quad g(\pi_i) \in (-\infty, \infty)$

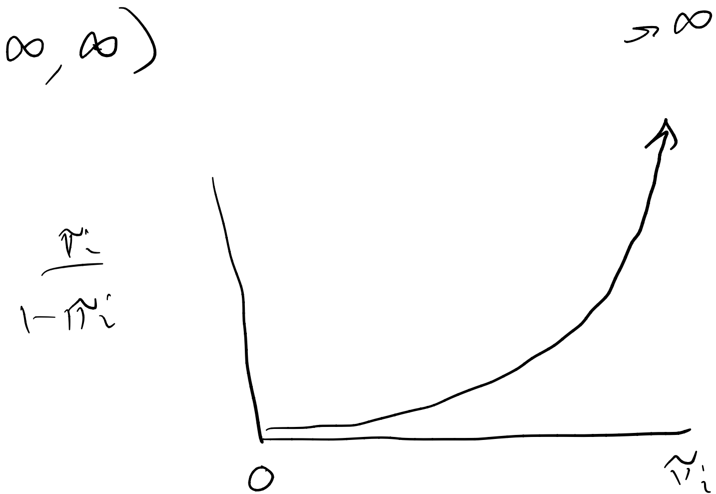
$\pi_i \in [0, 1]$

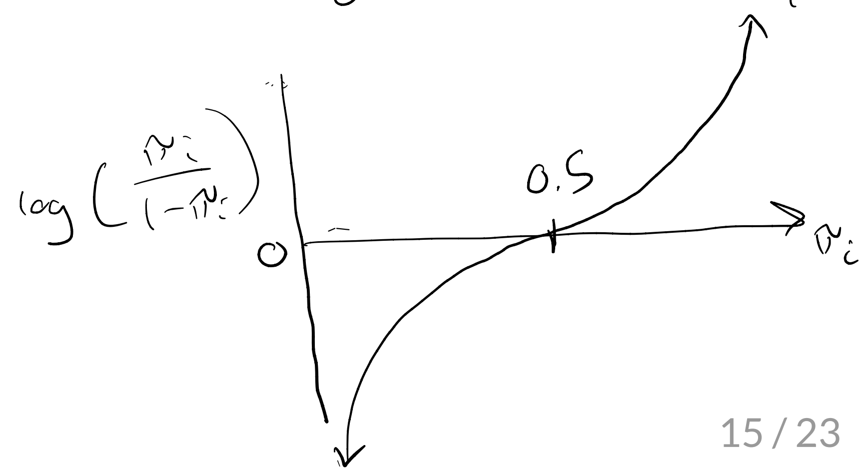
$\frac{\pi_i}{1-\pi_i} \in [0, \infty)$

$\log\left(\frac{\pi_i}{1-\pi_i}\right) \in (-\infty, \infty)$

$\uparrow$ natural log (aka ln)

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 Age_i$$





Logistic regression model

Y_i = dengue status (0 = negative, 1 = positive)

Age_i = age (in years)

Random component: $Y_i \sim Bernoulli(\pi_i)$

Systematic component: $\log\left(\dfrac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 \, Age_i$

why no ε_i??

ε_i in linear regression captures variability in Y_i

in logistic regression, $Bernoulli(\pi_i)$ captures variability / randomness in Y_i

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linear: $Y_i \sim N(\mu_i, \sigma^2)$

$\mu_i = \beta_0 + \beta_1 X_i$ ← no ε_i!

Class activity

- Work with a neighbor on the class activity (handout)
- I will collect your work at the end of class

For next time, read sections 6.4 and 6.6 in the textbook

Activity

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

Does the probability of dengue increase or decrease with age?

Activity

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

What are the predicted odds of dengue for a 5 year old patient?

Activity

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

For a 5 year old patient: is the predicted *probability* of dengue > 0.5, < 0.5, or $= 0.5$?